Lecture 16

Connections and geodesics

Objectives. *Differentiating a tangent field involves both its components and its coordinate basis. A connection accounts for this change. A metric selects one connection by compatibility and zero torsion. Its parallel transport preserves inner products, and its geodesics have parallel velocity.*

By the end of this lecture you will be able to:

- Construct the Levi-Civita connection from a nondegenerate metric.
- Compute parallel transport and geodesic equations in coordinates.
- Find a conserved geodesic quantity from a Killing field.
- Explain why the exponential is onto for a compact connected Lie group.

16.1 Affine connections

Let M be a smooth n-manifold without boundary and let $\mathfrak{X}(M)$ denote its smooth vector fields, as in Lecture 9. An *affine connection* is an $\mathbb{R}$-bilinear map $\nabla\colon \mathfrak{X}(M)\times\mathfrak{X}(M)\to\mathfrak{X}(M)$ that satisfies, for $f\in C^\infty(M)$,

$$\nabla_{fX}Y = f\nabla_X Y, \qquad \nabla_X(fY) = X(f)Y + f\nabla_X Y.$$

The field $\nabla_X Y$ is the *covariant derivative* of Y in direction X. The second rule prevents this operation from being bilinear over functions. For example, on Cartesian $\mathbb{R}^n$, the usual directional derivative $\nabla_X Y = X^i\partial_i Y^j\partial_j$ satisfies both rules.

The rules make the derivative local. If Y vanishes near p, choose a smooth cutoff f supported there with $f=1$ near p. Then $fY=0$, and the Leibniz rule at p gives $\nabla_X Y|_p = 0$. The first rule proves the same locality in X. Thus local fields can be extended with cutoffs before taking their derivative.

In coordinates $x^1,\dots,x^n$, write $\partial_i = \partial/\partial x^i$. The *Christoffel symbols* Γ_{ij}^k are defined by

$$\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k\partial_k, \qquad (\nabla_X Y)^k = X^i(\partial_i Y^k + \Gamma_{ij}^k Y^j).$$

Repeated indices range from 1 to n. The second formula follows from the two connection rules and shows that the symbols determine the connection. In a chart, this expression uses only X_p, Y_p and the first derivatives of Y at p.

The *torsion* is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$. It is a tensor: the derivative term in $\nabla_Y(fX) = Y(f)X + f\nabla_Y X$ cancels the derivative term in $[fX,Y] = f[X,Y] - Y(f)X$. Thus $T(fX,Y) = fT(X,Y)$. Antisymmetry gives linearity over functions in the other slot. To see why function-linearity gives a pointwise tensor, restrict locally using cutoffs. Expanding $Z = Z^i\partial_i$ in any function-linear slot gives $A(Z)(p) = Z^i(p)A(\partial_i)(p)$, which depends only on Z_p.

The connection is *torsion-free* when $T = 0$. Since coordinate fields commute, this is equivalent to $\Gamma^k_{ij} = \Gamma^k_{ji}$.

For a pseudo-Riemannian metric g, a *metric connection* satisfies

$$Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \qquad (X,Y,Z \in \mathfrak{X}(M)).$$

This identity differentiates the inner product using the connection.

16.2 The Levi-Civita connection

Theorem 16.1 (Levi-Civita). *Every pseudo-Riemannian manifold has a unique torsion-free metric connection. It is called the* Levi-Civita connection. *It satisfies the Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) \\ & - g(X,[Y,Z]) + g(Y,[Z,X]) + g(Z,[X,Y]). \end{aligned}$$

Its coordinate symbols are

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell}(\partial_i g_{\ell j} + \partial_j g_{\ell i} - \partial_\ell g_{ij}).$$

Proof. For uniqueness, write metric compatibility in the directions X, Y and Z. Add the first two identities and subtract the third. Replace $\nabla_Y X$ by $\nabla_X Y - [X,Y]$, and do the same for the other reversed pairs. The connection terms that remain are $2g(\nabla_X Y, Z)$, giving Koszul's formula. Nondegeneracy determines $\nabla_X Y$ uniquely.

For existence, denote the right side by $\mathcal{K}(X,Y,Z)$. Expand derivatives by the product rule and brackets by $[X, fY] = X(f)Y + f[X,Y]$. The derivative terms cancel to give

$$\begin{aligned} \mathcal{K}(X,Y,fZ) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(fX,Y,Z) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(X,fY,Z) &= f\mathcal{K}(X,Y,Z) + 2X(f)g(Y,Z). \end{aligned}$$

For example, in the first identity the extra terms are $X(f)g(Y,Z) + Y(f)g(X,Z) - Y(f)g(X,Z) - X(f)g(Y,Z) = 0$. The cutoff and coordinate-expansion argument of § 16.1 therefore makes $\mathcal{K}(X,Y,\cdot)/2$ a smooth covector field. Apply the inverse metric to define $\nabla_X Y$. The remaining identities give linearity in X and the Leibniz rule in Y. Additivity and real bilinearity follow directly from the formula.

Finally, exchanging the indicated slots in $\mathcal{K}$ gives

$$\begin{aligned} \mathcal{K}(X,Y,Z) - \mathcal{K}(Y,X,Z) &= 2g([X,Y],Z), \\ \mathcal{K}(X,Y,Z) + \mathcal{K}(X,Z,Y) &= 2Xg(Y,Z). \end{aligned}$$

The first identity proves zero torsion by nondegeneracy. The second proves metric compatibility. Substituting coordinate fields in Koszul, whose brackets vanish, gives $2g_{k\ell}\Gamma^k_{ij} = \partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}$. Multiplication by the inverse metric yields the stated symbols. ∎

For a constant metric matrix the symbols vanish in those coordinates. In particular, Minkowski geodesics will be straight lines. For an embedded surface in Euclidean three-space, the connection is $\nabla_X Y = \mathrm{pr}_{TM}(D_X Y)$, the tangential projection of the ambient derivative. Indeed, projection preserves the derivative rules. The ambient product rule gives metric compatibility because normal components pair to zero with tangent fields. The ambient identity $D_X Y - D_Y X = [X,Y]$ gives zero torsion. Uniqueness identifies this connection.

16.3 Parallel transport and geodesics

A *field along a curve* $\gamma\colon [a,b] \to M$ assigns $V(t) \in T_{\gamma(t)}M$ smoothly. In a coordinate segment its derivative is

$$\frac{DV}{dt} = (\dot V^k + \Gamma^k_{ij}(\gamma(t))\dot x^i V^j)\partial_k.$$

Equivalently, define this expression as $\dot v^j \partial_j + v^j \nabla_{\dot\gamma}\partial_j$ for $V = v^j\partial_j$ along the curve. In its first argument, the connection uses only the value of the direction field at the point. Write a local smooth basis as a row $e = (e_1, \dots, e_n)$, called a *frame*. For a new frame $e' = eA$ and coefficient columns $v = Av'$, define C by $\nabla_{\dot\gamma} e = eC$. The Leibniz rule and the coefficient change give

$$\nabla_{\dot\gamma}(eA) = e(\dot A + CA),$$
$$e(A\dot v' + \dot A v' + CAv') = e(\dot v + Cv).$$

Here $\dot A$ is the derivative along γ; the Leibniz rule supplies it. Thus the derivative is independent of the frame, including at repeated points of the curve. For $V = Y \circ \gamma$ it equals $\nabla_{\dot\gamma} Y$. A field is *parallel* when $DV/dt = 0$.

Proposition 16.2 (Parallel transport · *proved given linear ODE existence and uniqueness*)**.** *For a smooth curve $\gamma\colon [a,b] \to M$ and $v \in T_{\gamma(a)}M$, there is a unique parallel field with $V(a) = v$. The map $P_\gamma(v) = V(b)$ is a linear isomorphism. For a metric connection it preserves the metric.*

Proof. The coordinate equation is a linear ODE $\dot V = -C(t)V$, where $C^k{}_j = \Gamma^k_{ij}\dot x^i$. On a compact coordinate segment its smooth coefficients are bounded, so linear ODE existence and uniqueness give a solution throughout that segment. A finite subdivision places each closed segment in a chart. Propagate the initial value through the subdivision. Coordinate independence and uniqueness glue the solutions. The same uniqueness proves linearity. Reversing the curve gives the inverse map. In coordinates, metric compatibility reads $\partial_i g_{k\ell} = \Gamma^m_{ik} g_{m\ell} + \Gamma^m_{i\ell} g_{km}$. Differentiate $g_{k\ell}V^kW^\ell$ along the curve and substitute this identity. It gives the along-curve product rule. For parallel fields,

$$\frac{d}{dt} g(V,W) = g(DV/dt, W) + g(V, DW/dt) = 0.$$

Thus endpoint inner products agree. ∎

This is the ODE input already used in Theorem 9.1. The polar-coordinate figure shows that parallel fields can have changing components.

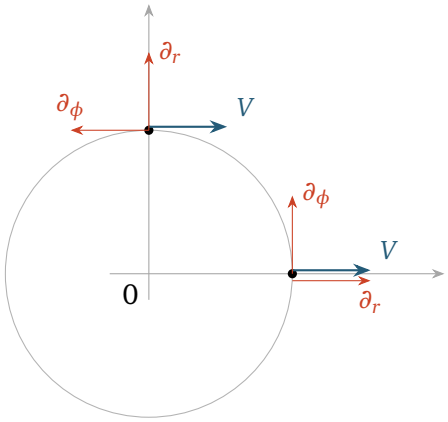


Figure 16.1: The same Cartesian vector $(1,0)$ at two points of the unit circle. Its polar components change from $(1,0)$ at $(1,0)$ to $(0,-1)$ at $(0,1)$. The arrows ∂_r and ∂_ϕ show the changing basis.

A *geodesic* is a parametrized curve with $D\dot\gamma/dt = 0$. Its coordinate equation is

$$\ddot x^k + \Gamma^k_{ij}\dot x^i \dot x^j = 0.$$

The local second-order ODE theorem gives a unique maximal geodesic γ_v with $\gamma_v(0) = p$ and $\dot\gamma_v(0) = v \in T_pM$. Here *maximal* means that its interval of definition admits no extension. The geodesic equation is a first-order field on TM, with bundle components $(\dot x^k, \dot v^k) = (v^k, -\Gamma^k_{ij} v^i v^j)$. The open smooth flow domain in Theorem 9.1 makes the *connection exponential* a smooth map $\exp_p \colon \mathcal{D}_p \to M$, given by $\exp_p(v) = \gamma_v(1)$. Its domain $\mathcal{D}_p \subset T_pM$ consists of the initial velocities for which the geodesic exists through time 1. The domain is open as the time-1 slice of that flow domain, restricted to T_pM. It contains zero because the constant curve is a geodesic. The connection is *geodesically complete* if every maximal geodesic is defined on all of $\mathbb{R}$; then $\mathcal{D}_p = T_pM$. For the Levi-Civita connection, $g(\dot\gamma, \dot\gamma)$ is constant by metric compatibility. In the Riemannian case, its square root is the speed.

The *energy* of a curve on $[a,b]$ is

$$E(\gamma) = \frac{1}{2}\int_a^b g(\dot\gamma,\dot\gamma)\,dt.$$

A curve is *stationary* when $\frac{d}{ds}\big|_0 E(\gamma_s) = 0$ for every smooth family γ_s with $\gamma_0 = \gamma$ and fixed endpoints.

Proposition 16.3 (Energy equations · *sketched; coordinate equivalence proved*)**.** *The Levi-Civita geodesics are precisely the stationary curves of the energy with fixed endpoints.*

The Euler–Lagrange equation, recalled from *Theoretical Mechanics*, Lecture 2, *Lagrange's equations*, is $d(\partial L/\partial \dot x^k)/dt - \partial L/\partial x^k = 0$. For $L = \frac12 g_{ij}\dot x^i \dot x^j$, its left side is

$$g_{k\ell}\ddot x^\ell + \partial_i g_{kj}\dot x^i \dot x^j - \frac12 \partial_k g_{ij}\dot x^i \dot x^j = g_{k\ell}(\ddot x^\ell + \Gamma^\ell_{ij}\dot x^i \dot x^j).$$

Nondegeneracy gives the geodesic equation. The variational step differentiates a fixed-endpoint variation and integrates its velocity term by parts. The endpoint term vanishes; the remaining integral pairs the displayed expression with the variation field. Sheet 16 supplies those details. No minimizing assertion follows in indefinite signature.

On the unit sphere, tangential projection gives $D\dot\gamma/dt = \ddot\gamma + |\dot\gamma|^2\gamma$. Indeed, differentiating $\gamma\cdot\gamma = 1$ twice gives $\gamma\cdot\ddot\gamma = -|\dot\gamma|^2$. For $v \in T_pS^2$ with $c = |v| > 0$, the unique solution is

$$\gamma_v(t) = \cos(ct)p + \frac{\sin(ct)}{c}v.$$

It lies on the sphere, has the prescribed initial data, and satisfies $\ddot\gamma = -c^2\gamma$. Its image is a *great circle*, the intersection with a plane through the origin. For $v = 0$ the geodesic is constant. In Minkowski coordinates, the zero Christoffel symbols instead give $\gamma_v(t) = p + tv$ for every initial vector, including zero and *null vectors*, which satisfy $\eta(v,v) = 0$.

16.4 Killing fields and conserved quantities

For a field K with local flow φ_t, define the metric Lie derivative by $L_Kg = \frac{d}{dt}\big|_0 \varphi_t^*g$. A *Killing field* satisfies $L_Kg = 0$. The flow law gives $\frac{d}{dt}\varphi_t^*g = \varphi_t^*(L_Kg)$, as for forms in Lecture 11. Thus the local flow of a Killing field consists of isometries, and conversely. For local fields Y and Z, put $Y_t = (\varphi_{-t})_*Y$ and $Z_t = (\varphi_{-t})_*Z$. The definition of tensor pullback gives $g(Y,Z)\circ\varphi_t = (\varphi_t^*g)(Y_t,Z_t)$. Differentiate at zero. The product rule and Theorem 9.3 give $Kg(Y,Z) = (L_Kg)(Y,Z) + g([K,Y],Z) + g(Y,[K,Z])$. Rearranging proves $(L_Kg)(Y,Z) = Kg(Y,Z) - g([K,Y],Z) - g(Y,[K,Z])$. Metric compatibility and zero torsion turn this into

$$(L_Kg)(Y,Z) = g(\nabla_YK,Z) + g(Y,\nabla_ZK).$$

Hence K is Killing exactly when the bilinear form $(Y,Z)\mapsto g(\nabla_YK,Z)$ is antisymmetric.

Proposition 16.4 (Killing conservation)**.** *Along a Levi-Civita geodesic γ, every Killing field K gives the constant quantity $g(K,\dot\gamma)$.*

Proof. Metric compatibility and the geodesic equation give

$$\frac{d}{dt}g(K,\dot\gamma) = g(\nabla_{\dot\gamma}K,\dot\gamma) = \frac{1}{2}(L_Kg)(\dot\gamma,\dot\gamma) = 0.$$ ∎

For example, a constant translation field in Minkowski space is Killing because the metric is constant. Its pairing with $\dot\gamma = v$ is constant, as the straight-line solution also shows.

16.5 Bi-invariant metrics and the group exponential

On a Lie group G, write $\ell_h(g) = hg$ and $r_h(g) = gh$. A metric is *bi-invariant* if every left and right translation is an isometry. Recall from Lecture 9 that each $X \in \mathfrak{g} = T_eG$ defines a complete left-invariant field X^L, whose identity curve $c_X(t)$ is a one-parameter subgroup. The *group exponential* $\exp\colon \mathfrak{g}\to G$ is given by $\exp X = c_X(1)$. Scaling the ODE gives $c_X(t) = \exp(tX)$, and the flow of X^L is $r_{\exp(tX)}$. This map differs in definition from Exp_p.

Theorem 16.5 (Bi-invariant geodesics · *proved given smooth ODE uniqueness*)**.** *For a bi-invariant pseudo-Riemannian metric, $\nabla_{X^L}Y^L = \frac{1}{2}[X,Y]^L$. The geodesics through the identity are exactly $t\mapsto\exp(tX)$, and $\mathrm{Exp}_e = \exp$ on all of $\mathfrak{g}$.*

Proof. The inner products of left-invariant fields are constant. Right translations are isometries, so X^L is Killing. The metric Lie derivative identity above therefore gives $g([X^L,Y^L],Z^L) + g(Y^L,[X^L,Z^L]) = 0$. Using this identity in the three bracket terms of Koszul gives

$$2g(\nabla_{X^L}Y^L,Z^L) = g([X^L,Y^L],Z^L).$$

The fields Z^L span each tangent space, so nondegeneracy proves the formula. Along c_X, its velocity is X^L, hence its acceleration is $\frac{1}{2}[X,X]^L = 0$. Uniqueness of the geodesic ODE gives all identity geodesics. Completeness of X^L gives their full real domains and the exponential equality. ∎

For compact G, a bi-invariant Riemannian metric exists. Here is a construction using Lecture 12's integration. A left-translated basis of $\mathfrak{g}$ gives a global frame and a positive, left-invariant volume form μ, normalized to have integral 1. Let b be any positive inner product on $\mathfrak{g}$. Define Ad_h to be the differential at the identity of $g \mapsto hgh^{-1}$. Average by

$$B(X,Y) = \int_G b(\mathrm{Ad}_{g^{-1}} X, \mathrm{Ad}_{g^{-1}} Y)\, \mu(g).$$

This is positive definite: a nonzero vector has a strictly positive integrand. Conjugation composes, so substituting $g = hk$ and using left-invariance of volume gives $B(\mathrm{Ad}_h X, \mathrm{Ad}_h Y) = B(X,Y)$. Extend B by left translation to a smooth metric. The identity $r_h \circ \ell_g = \ell_{gh} \circ (g' \mapsto h^{-1}g'h)$ shows that right translation acts on left-translated components by $\mathrm{Ad}_{h^{-1}}$. The Ad-invariance of B proves that the metric is bi-invariant.

For a connected Riemannian manifold, define a curve's length by $\ell(\gamma) = \int_a^b \sqrt{g(\dot\gamma,\dot\gamma)}\, dt$. The distance is the infimum of lengths of piecewise smooth curves joining its endpoints. A *minimizing geodesic* attains that infimum. The cited Hopf–Rinow theorem states that completeness of this distance is equivalent to geodesic completeness, and that on a complete connected Riemannian manifold every two points are joined by a minimizing geodesic (Tu, Theorem 16.6, p. 131). The distance induces the manifold topology (Lee, Lemma 6.2, pp. 94–95), so compactness makes it complete. Reparametrize the joining geodesic to $[0,1]$ by a linear change of time. Thus a compact connected G has every element $g = \mathrm{Exp}_e v = \exp v$ for some $v \in \mathfrak{g}$. This surjectivity conclusion is proved given integration and Hopf–Rinow; it does not assert injectivity of the exponential.

16.6 The round sphere in coordinates

For the unit sphere, the spherical parametrization is

$$p(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \qquad 0 < \theta < \pi.$$

Here ϕ lies in an angular chart. Dot products of its coordinate tangents give

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}.$$

The figure compares a great circle with a latitude.

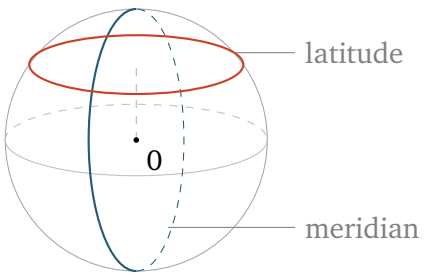


Figure 16.2: A meridian lies in a plane through the sphere's centre. A northern latitude lies in a plane parallel to the equator, away from the centre.

Example 16.6 (Sphere connection and angular momentum)**.** Only $\partial_\theta g_{\phi\phi} = 2\sin\theta\cos\theta$ is nonzero among the metric derivatives. The Christoffel formula gives

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2}\partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta,$$
$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \frac{1}{2\sin^2\theta}\partial_\theta g_{\phi\phi} = \cot\theta.$$

All other symbols vanish. Substitution gives

$$\ddot\theta - \sin\theta\cos\theta\,\dot\phi^2 = 0, \qquad \ddot\phi + 2\cot\theta\,\dot\theta\dot\phi = 0.$$

For a nonconstant latitude $\theta = \theta_0$ with constant angular speed, the first equation requires $\sin\theta_0\cos\theta_0 = 0$. Within this chart only $\theta_0 = \pi/2$ works: the equator is a geodesic. The coordinate poles are points, not nonconstant latitude curves.

The field $K = \partial_\phi$ generates rotations about the vertical axis, which preserve the metric. Its conserved quantity and its derivative are

$$g(K,\dot\gamma) = \sin^2\theta\,\dot\phi, \qquad \frac{d}{dt}(\sin^2\theta\,\dot\phi) = \sin^2\theta(\ddot\phi + 2\cot\theta\,\dot\theta\dot\phi) = 0.$$

This is the axial angular momentum for the unit-mass energy Lagrangian. The coordinate singularity at a pole does not make the smooth connection singular.

A geodesic need not minimize length on its entire interval. The following great-circle arcs have the same endpoints.

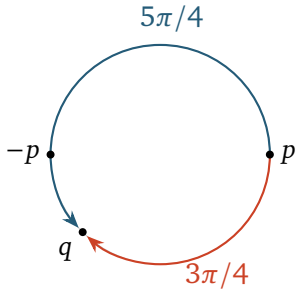


Figure 16.3: From p to q, the increasing-angle unit-circle arc has length $5\pi/4$. The decreasing-angle arc has length $3\pi/4$. The longer geodesic has passed the antipode $-p$.

The longer arc is a geodesic but cannot minimize length because the shorter arc joins the same endpoints. No local geodesic equation implies a global minimum.

Self-check. Compute the plane's Levi-Civita symbols in polar coordinates $r > 0$. Compare with Cartesian coordinates to test whether Christoffel symbols are a tensor.

Answer. The metric is $g = \mathrm{d}r\otimes\mathrm{d}r + r^2\mathrm{d}\phi\otimes\mathrm{d}\phi$. Its only nonzero derivative is $\partial_r g_{\phi\phi} = 2r$. The formula gives

$$\Gamma^r_{\phi\phi} = -r, \qquad \Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r.$$

All other symbols vanish. The Cartesian symbols are zero, whereas these are nonzero at the same points. A tensor with zero components in one basis has zero components in every basis. Thus Christoffel symbols are not tensor components.

Summary. A nondegenerate metric determines the torsion-free metric connection by Koszul's formula. Parallel transport solves a linear ODE and preserves the metric. Geodesics have parallel velocity and constant metric norm. Killing fields give conserved pairings with that velocity. For a bi-invariant metric the identity geodesics are one-parameter subgroups. Compact averaging and the cited Hopf–Rinow theorem make the group exponential onto for compact connected groups. Lecture 17 computes curvature from this connection.

Reading. Tu, *Differential Geometry*, §6, §13–§15, especially Problems 15.4–15.5, and Theorem 16.6, p. 131. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Theorem 5.4, pp. 68–70, and Lemma 6.2, pp. 94–95. Nakahara, §7.2, §7.4 and §7.7.